

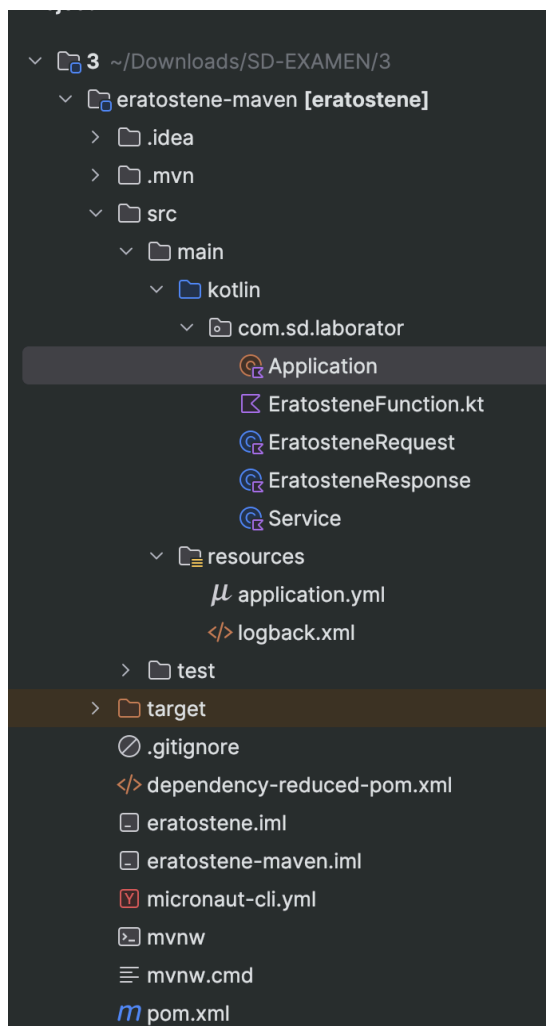
Pomind de la aplicatia exemplu pentru ciurul lui Eratostene din laboratorul 11, sa se modifice aceasta astfel incat sa se calculeze $bn = \sum_{i=1}^n (ai * ai)$, $i=1..n$. Apoi utilizând această funcție în programul principal să se calculeze pornind de la 100 de valori initializate aleator într-un ADT A si apoi să se depună într-un ADT B rezultatele pentru $n=1..100$.

NOTITE

1. Mergi pe site-ul asta: <https://micronaut.io/launch> si generezi proiectul dupa setarile astea

Application Type Command Line Application	Java Version 21	Name find-numbers	Base Package com.example
Micronaut Version ☒ 4.8.3 ☐ 4.8.4-SNAPSHOT ☐ 3.10.1	Language ☐ Java ☐ Groovy ☒ Kotlin	Build Tool ☐ Gradle ☐ Gradle Kotlin ☒ Maven	Test Framework ☒ JUnit ☐ Spock ☐ Kotest
+ FEATURES	- DIFF	🔍 PREVIEW	🔑 GENERATE PROJECT

2. Dezarhiveaza
3. Modifica Pom.xml-ul dupa cum vezi mai jos



Application

```
package com.sd.laborator

import io.micronaut.runtime.Micronaut

object Application {

    @JvmStatic
    fun main(args: Array<String>) {
        Micronaut.build()
            .packages("com.sd.laborator")
            .mainClass(Application::javaClass)
            .start()
    }
}
```

ErastoteneFunction

```
package com.sd.laborator

import io.micronaut.function.FunctionBean
import io.micronaut.function.executor.FunctionInitializer
import org.slf4j.Logger
import org.slf4j.LoggerFactory
import java.util.function.Function
import javax.inject.Inject

// echo '{"number": "8"}' | java -jar target/eratostene-0.1.jar

@FunctionBean("eratostene")
class EratosteneFunction :
    FunctionInitializer(),
    Function<EratosteneRequest, EratosteneResponse> {
    @Inject
    private lateinit var service: Service

    private val LOG: Logger =
        LoggerFactory.getLogger(EratosteneFunction::class.java)

    override fun apply(request: EratosteneRequest): EratosteneResponse {
        // preluare numar din parametrul de intrare al functiei
        val seed = request.getNumber()

        val response = EratosteneResponse()

        // se face calculul si se seteaza proprietatile pe obiectul cu
        rezultatul
        response.setResponse(service.calcul(seed.toLong()))
    }
}
```

```

        response.setMessage("Calcul efectuat cu succes!")

        LOG.info("Calcul incheiat!")

        return response
    }
}

fun main(args: Array<String>) {
    val function = EratosteneFunction()
    function.run(args, { context ->
function.apply(context.get(EratosteneRequest::class.java)) })
}

```

EratosteneRequest

```

package com.sd.laborator

import io.micronaut.core.annotation.Introspected

@Introspected
class EratosteneRequest {
    private lateinit var number: Integer

    fun getNumber(): Int = number.toInt()
}

```

EratosteneResponse

```

package com.sd.laborator

import io.micronaut.core.annotation.Introspected

@Introspected
class EratosteneResponse {
    private var message: String? = null
    private var response: List<Int>? = null

    fun setResponse(primes: List<Int>?) {
        this.response = primes
    }

    fun setMessage(message: String?) {

```

```

        this.message = message
    }

    fun getMessage(): String?
    {
        return this.message
    }

    fun getResponse(): List<Int>?
    {
        return this.response
    }
}

```

Service

```
@file:Suppress("ktlint:standard:no-wildcard-imports")
```

```

package com.sd.laborator

import java.util.*
import javax.inject.Singleton

@Singleton
class Service {
    @Suppress("ktlint:standard:property-naming")
    private val random_vect = Vector<Int>(100)

    fun calcul(seed: Long): List<Int> {
        val adt = Vector<Int>(100)

        for (i in 0 until 99) {
            random_vect.add((Math.random() * 100).toInt())
            adt.add(random_vect[i] * random_vect[i])
        }

        return adt
    }
}

```

Pom.xml

```

<?xml version="1.0" encoding="UTF-8"?><project
xmlns="http://maven.apache.org/POM/4.0.0"

```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.sd.laborator</groupId>
  <artifactId>eratostene</artifactId>
  <version>0.1</version>
  <properties>
    <micronaut.version>1.3.4</micronaut.version>
    <jdk.version>1.8</jdk.version>
    <maven.compiler.target>${jdk.version}</maven.compiler.target>
    <maven.compiler.source>${jdk.version}</maven.compiler.source>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <project.reporting.outputEncoding>UTF-8</project.reporting.outputEncoding>
    <kotlinVersion>1.3.50</kotlinVersion>
    <maven-surefire-plugin.version>2.22.2</maven-surefire-plugin.version>
    <maven-failsafe-plugin.version>2.22.2</maven-failsafe-plugin.version>
    <exec.mainClass>io.micronaut.function.executor.Application</exec.mainClass>
  </properties>
  <repositories>
    <repository>
      <id>jcenter.bintray.com</id>
      <url>https://jcenter.bintray.com</url>
    </repository>
  </repositories>
  <dependencyManagement>
    <dependencies>
      <dependency>
        <groupId>io.micronaut</groupId>
        <artifactId>micronaut-bom</artifactId>
        <version>${micronaut.version}</version>
        <type>pom</type>
        <scope>import</scope>
      </dependency>
    </dependencies>
  </dependencyManagement>
  <dependencies>
    <dependency>
      <groupId>org.jetbrains.kotlin</groupId>
      <artifactId>kotlin-stdlib-jdk8</artifactId>
      <version>${kotlinVersion}</version>
      <scope>compile</scope>
    </dependency>
    <dependency>
      <groupId>org.jetbrains.kotlin</groupId>
      <artifactId>kotlin-reflect</artifactId>
      <version>${kotlinVersion}</version>
      <scope>compile</scope>
    </dependency>
  </dependencies>

```

```
<dependency>
  <groupId>io.micronaut</groupId>
  <artifactId>micronaut-runtime</artifactId>
  <scope>compile</scope>
</dependency>
<dependency>
  <groupId>io.micronaut</groupId>
  <artifactId>micronaut-function-aws</artifactId>
  <scope>compile</scope>
</dependency>
<dependency>
  <groupId>com.amazonaws</groupId>
  <artifactId>aws-lambda-java-log4j2</artifactId>
  <version>1.0.0</version>
  <scope>runtime</scope>
</dependency>
<dependency>
  <groupId>org.apache.logging.log4j</groupId>
  <artifactId>log4j-slf4j-impl</artifactId>
  <version>2.9.1</version>
  <scope>runtime</scope>
</dependency>
<dependency>
  <groupId>com.fasterxml.jackson.module</groupId>
  <artifactId>jackson-module-kotlin</artifactId>
  <version>2.9.8</version>
  <scope>runtime</scope>
</dependency>
<dependency>
  <groupId>io.micronaut.test</groupId>
  <artifactId>micronaut-test-kotlintest</artifactId>
  <scope>test</scope>
</dependency>
<dependency>
  <groupId>io.mockk</groupId>
  <artifactId>mockk</artifactId>
  <version>1.9.3</version>
  <scope>test</scope>
</dependency>
<dependency>
  <groupId>io.kotlintest</groupId>
  <artifactId>kotlintest-runner-junit5</artifactId>
  <version>3.3.2</version>
  <scope>test</scope>
</dependency>
<dependency>
  <groupId>org.apache.logging.log4j</groupId>
  <artifactId>log4j-api</artifactId>
  <version>2.13.2</version>
```

```

        <scope>runtime</scope>
    </dependency>
</dependencies>
<build>
    <plugins>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-shade-plugin</artifactId>
            <version>3.1.0</version>
            <configuration>
                <createDependencyReducedPom>>false</createDependencyReducedPom>
                <filters>
                    <filter>
                        <artifact>*:*</artifact>
                        <excludes>
                            <exclude>**/Log4j2Plugins.dat</exclude>
                        </excludes>
                    </filter>
                </filters>
            </configuration>
        </plugin>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-shade-plugin</artifactId>
            <version>3.1.0</version>
            <executions>
                <execution>
                    <phase>package</phase>
                    <goals>
                        <goal>shade</goal>
                    </goals>
                    <configuration>
                        <transformers>
                            <transformer
implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransfo
rmer">
                                <mainClass>${exec.mainClass}</mainClass>
                            </transformer>
                            <transformer
implementation="org.apache.maven.plugins.shade.resource.ServicesResourceTransfo
rmer"/>
                                </transformers>
                        </configuration>
                    </execution>
                </executions>
            </plugin>
            <plugin>
                <groupId>org.codehaus.mojo</groupId>
                <artifactId>exec-maven-plugin</artifactId>

```

```

<version>1.6.0</version>
<configuration>
  <executable>java</executable>
  <arguments>
    <argument>-classpath</argument>
    <classpath/>

    <argument>${exec.mainClass}</argument>
  </arguments>
</configuration>
</plugin>
<plugin>
  <artifactId>kotlin-maven-plugin</artifactId>
  <groupId>org.jetbrains.kotlin</groupId>
  <version>${kotlinVersion}</version>
  <configuration>
    <jvmTarget>${jdk.version}</jvmTarget>
    <compilerPlugins>
      <plugin>all-open</plugin>
    </compilerPlugins>
    <pluginOptions>
      <option>all-open:annotation=io.micronaut.aop.Around</option>
    </pluginOptions>
  </configuration>
  <executions>
    <execution>
      <id>kapt</id>
      <goals>
        <goal>kapt</goal>
      </goals>
      <configuration>
        <sourceDirs>
          <sourceDir>${project.basedir}/src/main/kotlin</sourceDir>
        </sourceDirs>
        <annotationProcessorPaths>
          <annotationProcessorPath>
            <groupId>io.micronaut</groupId>
            <artifactId>micronaut-inject-java</artifactId>
            <version>1.3.4</version>
          </annotationProcessorPath>
          <annotationProcessorPath>
            <groupId>io.micronaut</groupId>
            <artifactId>micronaut-validation</artifactId>
            <version>1.3.4</version>
          </annotationProcessorPath>
        </annotationProcessorPaths>
      </configuration>
    </execution>
    <execution>

```



```

        <id>compile</id>
        <goals>
            <goal>compile</goal>
        </goals>
        <configuration>
            <sourceDirs>
                <sourceDir>${project.basedir}/src/main/kotlin</sourceDir>
                <sourceDir>${project.basedir}/src/main/java</sourceDir>
            </sourceDirs>
        </configuration>
    </execution>
    <execution>
        <id>test-kapt</id>
        <goals>
            <goal>test-kapt</goal>
        </goals>
        <configuration>
            <sourceDirs>
                <sourceDir>${project.basedir}/src/test/kotlin</sourceDir>
            </sourceDirs>
            <annotationProcessorPaths>
                <annotationProcessorPath>
                    <groupId>io.micronaut</groupId>
                    <artifactId>micronaut-inject-java</artifactId>
                    <version>1.3.4</version>
                </annotationProcessorPath>
                <annotationProcessorPath>
                    <groupId>io.micronaut</groupId>
                    <artifactId>micronaut-validation</artifactId>
                    <version>1.3.4</version>
                </annotationProcessorPath>
            </annotationProcessorPaths>
        </configuration>
    </execution>
    <execution>
        <id>test-compile</id>
        <goals>
            <goal>test-compile</goal>
        </goals>
        <configuration>
            <sourceDirs>
                <sourceDir>${project.basedir}/src/test/kotlin</sourceDir>
                <sourceDir>${project.basedir}/target/generated-sources/kapt/test</sourceDir>
            </sourceDirs>
        </configuration>
    </execution>
</executions>
<dependencies>

```

```
        <dependency>
          <groupId>org.jetbrains.kotlin</groupId>
          <artifactId>kotlin-maven-allopen</artifactId>
          <version>${kotlinVersion}</version>
        </dependency>
      </dependencies>
    </plugin>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.7.0</version>
      <configuration>
        <proc>none</proc>
        <source>${jdk.version}</source>
        <target>${jdk.version}</target>
      </configuration>
      <executions>
        <execution>
          <id>default-compile</id>
          <phase>none</phase>
        </execution>
        <execution>
          <id>default-testCompile</id>
          <phase>none</phase>
        </execution>
        <execution>
          <id>java-compile</id>
          <phase>compile</phase>
          <goals>
            <goal>compile</goal>
          </goals>
        </execution>
        <execution>
          <id>java-test-compile</id>
          <phase>test-compile</phase>
          <goals>
            <goal>testCompile</goal>
          </goals>
        </execution>
      </executions>
    </plugin>
  </plugins>
  <pluginManagement>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-surefire-plugin</artifactId>
        <version>${maven-surefire-plugin.version}</version>
        <configuration>
```

```
        <includes>
            <include>**/*Spec.*</include>
            <include>**/*Test.*</include>
        </includes>
    </configuration>
    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter-engine</artifactId>
            <version>5.6.0</version>
        </dependency>
    </dependencies>
</plugin>
<plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-failsafe-plugin</artifactId>
    <version>${maven-failsafe-plugin.version}</version>
    <executions>
        <execution>
            <goals>
                <goal>integration-test</goal>
                <goal>verify</goal>
            </goals>
        </execution>
    </executions>
</plugin>
</plugins>
</pluginManagement>
</build>
</project>
```